

PRANAV MATHAM

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SUMMARY

Software engineer (BE, Computer Science, 2026) with 9+ months of hands-on experience across three software roles, building, testing, and debugging full-stack applications with Python, Django, React, and REST APIs in Agile/Scrum teams. IEEE-published researcher who writes clean, tested code and documents projects with READMEs covering setup, schema, and system flow.

EDUCATION

SJB Institute of Technology | BE, Computer Science and Engineering

Nov 2022 – Jun 2026

- Relevant Coursework: Software Engineering & Project Management, Data Structures, Analysis & Design of Algorithms, OOP (Java), DBMS, Complexity Analysis, AI, Machine Learning, Deep Learning.

SKILLS

Programming Languages: Python, Java, JavaScript, C++, C, SQL

Software Engineering: SDLC, Agile/Scrum, Unit Testing, Debugging, REST APIs, Authentication, READMEs, Git/GitHub

Frameworks & Tools: React, Node.js, Django, FastAPI, Flask, VS Code, PyCharm

Core CS: Data Structures & Algorithms, Object-Oriented Design, DBMS, Distributed Systems, Complexity Analysis

AI/Research: Machine Learning, Deep Learning, PyTorch, OpenCV, NumPy, Explainable AI

Publication: “A Novel Framework for Automated Detection and Classification of Thyroid Nodules,” IEEE, 2026

WORK EXPERIENCE

Software Intern — Vectora Ventures Private Limited, Bengaluru, India

Aug 2026 – Present

- Contribute across the software development lifecycle through coding, debugging, testing, and documentation; investigate and resolve software issues in a hybrid Agile/Scrum team.
- Key Skills: Software Development, Debugging, Testing, Documentation, Agile/Scrum.

Python Full-Stack Developer Intern — Test Yantra Software Solutions, Bengaluru, India

Jan 2026 – Jun 2026

- Designed and built a Python/Django/SQL backend for BusPass, modeling relational schemas for student profiles, routes, and pass applications to support an end-to-end user workflow with status tracking.
- Implemented REST/AJAX data flows, Django authentication (hashed passwords, student/admin roles), and admin approval workflows; debugged frontend-backend integration issues and validated application behavior through systematic testing.
- Key Skills: Python, Django, SQL, DBMS, REST APIs, Unit Testing, Git/GitHub.

Software Development Trainee — PYSPIDERS, Bengaluru, India

Dec 2025 – May 2026

- Built responsive, user-facing web components using HTML, CSS, JavaScript, and Python, applying structured development practices and systematic debugging across project builds.
- Collaborated in Agile workflows to diagnose and resolve frontend-backend integration issues across project builds.
- Key Skills: HTML, CSS, JavaScript, Python, UI Development, Debugging, Agile/Scrum.

RESEARCH & SOFTWARE PROJECTS

Urban-Tech (NeuroSphere-X) – Smart City Traffic Platform — [Urban-Tech Project](#)

- Engineered a real-time distributed platform using FastAPI, Python, WebSockets, PyTorch, React, and Leaflet to ingest and process telemetry, geospatial routing, and live dashboard events.
- Designed data-driven analytics pipelines with dynamic heatmaps and emergency-vehicle telemetry, emphasizing low-latency event processing and fault-tolerant data flow.

Railway Ticket Reservation Management System — [Railway Project](#)

- Built a Flutter/Dart reservation application using BLoC, REST/GraphQL APIs, Firebase, and Hive/SQLite for structured ticket-data handling and application workflows.
- Designed database-backed booking and synchronization workflows with local caching; validated performance and scalability through load and latency testing.

AI-Powered Nutrition Tracking Website – Nutri-AI — [Nutri-AI Project](#)

- Built a full-stack nutrition platform using React, Node.js, Flask, and REST APIs, integrating OpenCV, PIL, and NumPy for food-image processing with USDA nutrition data.

Thyroid Nodule Detection & Classification using Deep Learning — [Thyroid Project](#)

- Designed and evaluated an end-to-end ultrasound image-analysis pipeline using EfficientNet-B0 for preprocessing, segmentation, feature extraction, and classification, achieving ~95% model accuracy.
- Added Grad-CAM for explainable predictions, user authentication, and relational scan-history and confidence data; research findings published in IEEE, 2026.